

```

1 % Vehicle parameters
2 m = 1500;      % mass (kg)
3 Iz = 3000;     % yaw inertia (kg*m^2)
4
5 lf = 1.2;      % front axle distance (m)
6 lr = 1.6;      % rear axle distance (m)
7
8 Cf = 19000;    % front cornering stiffness (N/rad)
9 Cr = 33000;    % rear cornering stiffness (N/rad)
10
11 Vx = 15;       % constant forward speed (m/s)
12
13 A = [ 0, 1, 0, 0;
14       0, 0, 1, Vx;
15       0, 0, -(Cf+Cr)/(m*Vx), -(Cf*lf-Cr*lr)/(m*Vx) - Vx;
16       0, 0, -(Cf*lf-Cr*lr)/(Iz*Vx), -(Cf*lf^2+Cr*lr^2)/(Iz*Vx) ];
17
18 B = [0;
19      0;
20      Cf/m;
21      Cf*lf/Iz];
22
23 C = eye(4);     % output all states
24 D = zeros(4,1); % no direct input-output effect
25
26 sys = ss(A,B,C,D);
27
28 t = 0:0.01:5;
29 u = 0.05 * ones(size(t)); % constant steering input
30
31 lsim(sys, u, t)
32
33 mpcDesigner(sys)
34
35 Ts = 0.05;      % sample time (seconds)
36
37 sys_d = c2d(sys, Ts);
38
39 Ad = sys_d.A;
40 Bd = sys_d.B;
41 Cd = sys_d.C;
42 Dd = sys_d.D;
43
44 t = 0:Ts:5;
45 u = 0.05 * ones(size(t));
46
47 lsim(sys_d, u, t)
48
49 mpcobj = mpc(sys_d, Ts);
50
51 % Discretization
52 Ts = 0.05;
53 sys_d = c2d(sys, Ts);
54
55 % Check matrices
56 Ad = sys_d.A;
57 Bd = sys_d.B;
58
59 % Test
60 step(sys_d);
61

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62 % MPC
63 mpcDesigner(sys_d);
64
65 mpcobj = mpc(sys_d, Ts);
66
67 C = eye(4);
68
69 mpcobj.Model.Nominal.Y = [0; 0; 0; 0];
70
71 mpcobj.Weights.OutputVariables = [10 5 0 0];
72 mpcobj.Weights.ManipulatedVariables = 0.1;
73 mpcobj.Weights.ManipulatedVariablesRate = 1;
74
75 mpcobj.MV.Min = -0.5;    % radians (~ -28 deg)
76 mpcobj.MV.Max = 0.5;
77
78 mpcobj.MV.RateMin = -0.1;
79 mpcobj.MV.RateMax = 0.1;
80
81 mpcobj.PredictionHorizon = 20;
82 mpcobj.ControlHorizon = 5;
83
84 mpcDesigner(sys_d)
85
86 T = 50;
87 r = [0 0 0 0];    % reference (stay at zero error)
88
89 sim(mpcobj, T, r)
90
91 % Create MPC
92 mpcobj = mpc(sys_d, Ts);
93
94 % Weights
95 mpcobj.Weights.OutputVariables = [10 5 0 0];
96 mpcobj.Weights.ManipulatedVariables = 0.1;
97 mpcobj.Weights.ManipulatedVariablesRate = 1;
98
99 % Constraints
100 mpcobj.MV.Min = -0.5;
101 mpcobj.MV.Max = 0.5;
102
103 % Horizons
104 mpcobj.PredictionHorizon = 20;
105 mpcobj.ControlHorizon = 5;
106
107 % Launch GUI
108 mpcDesigner(sys_d);

```